

Edison Ricardo Altamirano Avila

Robotics, Embedded Systems, IC Design, AI, and VLSI Engineering
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Education

Stanford University

M.S. in Electrical Engineering

Jan 2024 – present

Stanford, CA

- Focus on mixed-signal ICs, VLSI systems, and sensor hardware/software integration.
- Coursework includes analog IC design, VLSI implementation, and radar-oriented sensing systems.

Instituto Tecnológico y de Estudios Superiores de Monterrey ITESM

B.S. in Digital Systems and Robotics Engineering

Aug 2018 – June 2023

Monterrey, Mexico

- Built robotics and autonomous-systems depth through Roborregos, VANTTEC, and Smart Factory programs.

Experience

Application Engineer

ZF Group

Sept 2023 – Sept 2025

Monterrey, Mexico

- Designed AUTOSAR Classic software architecture for commercial vehicle braking systems.
- Modeled braking physics in MATLAB/Simulink to support Software-in-the-Loop validation.
- Applied MISRA-C, VectorCAST, and DaVinci Developer to verification and reliability workflows.

CTO

CHAKRI

Aug 2020 – Sept 2025

Quito, Ecuador

- Co-founded product efforts connecting farmers, vendors, and customers through integrated digital platforms.
- Delivered BAC Modelos as a scalable Next.js + Firebase + Elasticsearch platform for a Mexico-based client.
- Built secure downloads, search APIs, manufacturer onboarding, analytics, and admin workflows.

Autonomous Vehicle Intern

ZF Group

Aug 2022 – June 2023

Monterrey, Mexico

- Integrated ROS 2, CAN bus control, and PWM acceleration / brake modulation.
- Implemented visual SLAM and camera + LiDAR sensor fusion for autonomous operation.
- Used simulation before deployment to validate CAN-based behavior and integration.

Supply Chain Management Intern

John Deere

Jan 2022 – Jan 2023

Mexico

- Built IM Reserver with SAP ABAP-driven inventory integration and reservation workflows.
- Used Azure Functions, Azure Logic Apps, AWS Lambda, and Power Platform for backend logic and dashboards.
- Developed ShieldGuard workflows using OpenCV/OpenVINO to track hand-item interactions for theft prevention.

Software Developer

RoBorregos

Oct 2022 – June 2023

Monterrey, Mexico

- Used ROS Noetic, OpenCV 4.2, PCL, MoveIt!, TensorFlow 1.15, C++ , and Python.
- Built robust state-machine, navigation, and task-execution flows for service-robot scenarios.

Lead Software Engineer

Vanttec

Aug 2020 – June 2023

Monterrey, Mexico

- Led software for autonomous surface and underwater vehicles in RoboBoat and RoboSub.
- Used ROS/ROS2, ORBSLAM3, vision pipelines, radio communication, and CAN-based vehicle control.

Research Assistant

Tecnológico de Monterrey

Jan 2019 – Aug 2022

Monterrey, Mexico

- Helped build automated factory systems with AGVs, PLC-connected workflows, digital twins, and VR interfaces.
- Contributed to IEEE EDUCON, EDUNINE, and IALF work on digital twins and robotics-skills acquisition.
- Supported industrial training and teaching on robotics, monitoring, and control workflows.

Projects [see full list](#)

Arbaban Lab Radar & Multimodal Sensing

2024-present

- Conduct research in ambient sensing and multimodal sensor fusion for autonomous and intelligent systems.
- Integrate TI IWR6843 70 GHz radar, Ouster LiDAR, and ZED, ELP, and Orbbec RGB-D cameras into synchronized perception pipelines.
- Build research depth at the boundary of sensing hardware, perception, and intelligent systems.

EE372 Analog IC Tapeout Track

Jan 2025

- Covered schematic capture, physical layout, DRC/LVS, and fabrication-oriented signoff steps.
- Built tapeout awareness instead of only simulated circuit experience.

EE271 / EE272 DNN Accelerator on SKY130

2024-2025

- Used Verilog/SystemVerilog, Cadence Genus, Cadence Innovus, and signoff workflows.
- Completed DNN accelerator implementation targeting tapeout-ready GDS output.

EE233 FM Superheterodyne Radio Receiver

Jan 2025

- Designed a complete 87.5-108 MHz FM receiver using a Si5351 PLL IC, I/Q quadrature demodulation, and Raspberry Pi Pico control.

Skills

Semiconductor & Sensing Hardware: Mixed-signal ICs, analog front-ends, radar sensing, multimodal sensing, RTL-to-GDS flow, physical verification

Robotics & Autonomy: ROS/ROS2, navigation, visual SLAM, ORBSLAM3, sensor fusion, MoveIt!, mission logic, simulation-to-deployment integration

Embedded & Automotive: AUTOSAR Classic, CAN bus, PWM control, MATLAB/Simulink, MISRA-C, VectorCAST, DaVinci Developer

AI / Perception: OpenCV, PCL, TensorFlow, OpenVINO, robot perception, grasping, multimodal AI, applied vision systems

Programming: Python, C++ , C, Verilog/SystemVerilog, TypeScript, Flutter

Publications

Visual Navigation Algorithms for Mobile Manipulators in Service Shops., *International Conference on Robotics and Automation* Jan 2023

Rodriguez-Padilla, C., Ahmad, R., **Altamirano-Avila, E.**

A Digital Twin Implementation for Mobile and Collaborative Robot Scenarios for Teaching Robotics Jan 2022
Based on Robot Operating System., *IEEE EDUCON*

Altamirano-Avila, E., Ahmad, R., Rodriguez-Padilla, C.

Towards a Mixed Virtual Reality Environment Implementation to Enable Industrial Robot Program-Jan 2023
ming Competencies within a Cyber-Physical Factory., *IEEE EDUCON*

Vazquez-Hurtado, C., **Altamirano-Avila, E.**, Ahmad, R.

Awards

- Most Consistent Solution, AIRLab Stacking Challenge. Humanoid robot tournament hosted in the Netherlands. [project page](#)
- 1st place RoboBoat and 3rd place RoboSub with VANTTEC. Autonomous boat and submarine tournaments hosted in the US. [project page](#)